

Hamming Schemes: Wreath-Product Permutation Formula Proof and Exact Verification

Executable companion note

17 July 2026

Abstract

We prove and test the generalized Molien formula for the full Hamming graph automorphism group $S_q \wr S_D = S_q^D \rtimes S_D$ acting on Q^D . The tests construct every wreath-product element, not merely pure coordinate permutations, and compare matrix, cycle, and orbit calculations through total degree three.

1 Permutation Molien and orbit theorem

For any finite action $G \curvearrowright X$ with vertex-cycle counts $c_\ell(g)$,

$$\mathcal{S}_{G,X}(\mathbf{t}) = \frac{1}{|G|} \sum_{g \in G} \prod_{i,\ell} (1 - t_i^\ell)^{-c_\ell(g)}. \quad (1)$$

This follows because an ℓ -cycle block satisfies $\det(I - tP_\ell) = 1 - t^\ell$. Moreover,

$$[m_\tau] \mathcal{S}_{G,X} = \# \left(G \setminus \prod_j \text{Sym}^{\tau_j}(X) \right), \quad (2)$$

since $\text{Sym}^d(\mathbb{C}[X])$ has its monomial basis indexed by d -multisets of vertices. This gives a proof route independent of determinant algebra.

2 Fixed words in the full wreath product

Write $g = (\sigma_1, \dots, \sigma_D; \pi)$. For a fixed power g^r , follow a coordinate around a cycle $C = (i_1, \dots, i_s)$ of its coordinate permutation. The successive alphabet relabelings compose to a permutation $h_{C,r} \in S_q$. A fixed word is determined on C by one symbol fixed by $h_{C,r}$, and different coordinate cycles are independent. Therefore

$$F_r^H(g) = \prod_{C \in \text{Cyc}(\pi^r)} |\text{Fix}_Q(h_{C,r})|. \quad (3)$$

The exact vertex-cycle counts are then

$$c_\ell^H(g) = \frac{1}{\ell} \sum_{d|\ell} \mu(\ell/d) F_d^H(g). \quad (4)$$

Substitution in (1) proves the proposed Hamming formula. For pure coordinate permutations the label products are identities, reducing (3) to $q^{c(\pi^r)}$.

Two ordered pairs of words are equivalent exactly when they have equal Hamming distance. Hence vertex transitivity and the distances $0, \dots, D$ give

$$[m_{(1)}] = 1, \quad [m_{(2)}] = [m_{(1,1)}] = D + 1.$$

3 Independent computational checks

The implementation explicitly constructs every word permutation and its $q^D \times q^D$ matrix P . It computes complete symmetric characters from actual matrix traces,

$$dh_d(P) = \sum_{r=1}^d \text{tr}(P^r) h_{d-r}(P), \quad h_0 = 1,$$

and averages $\prod_j h_{\tau_j}(P)$. A second route extracts coefficients from (1); a third explicitly traverses the orbits in (2). Separately, the coordinate maps and alphabet labels are powered by composition, (3) is evaluated, and (4) is required to reproduce the actual vertex cycles for every group element.

4 Exact results

Values are ordered by $\tau = (1), (2), (1, 1), (3), (2, 1), (1, 1, 1)$.

Action	$ G $	$ X $	common coefficient vector
$S_2 \wr S_2$ on Q^2	8	4	(1, 3, 3, 4, 7, 10)
$S_2 \wr S_3$ on Q^3	48	8	(1, 4, 4, 7, 13, 20)
$S_3 \wr S_2$ on Q^2	72	9	(1, 3, 3, 7, 11, 15)

All 18 matrix/cycle/orbit comparisons and every fixed-word reconstruction pass. For $(q, D) = (3, 2)$ at $(t_1, t_2) = (0.037, 0.061)$, the determinant and cycle averages are 1.1254339873575003 and 1.1254339873575000; the absolute error is 2.22×10^{-16} .

5 Reproduction

Run

```
marimo run for_this_guy/association_scheme_permutation_representations/
    hamming/notebook.py
```

from the repository root. The table in the notebook exposes every exact comparison and the global fixed-point reconstruction result.